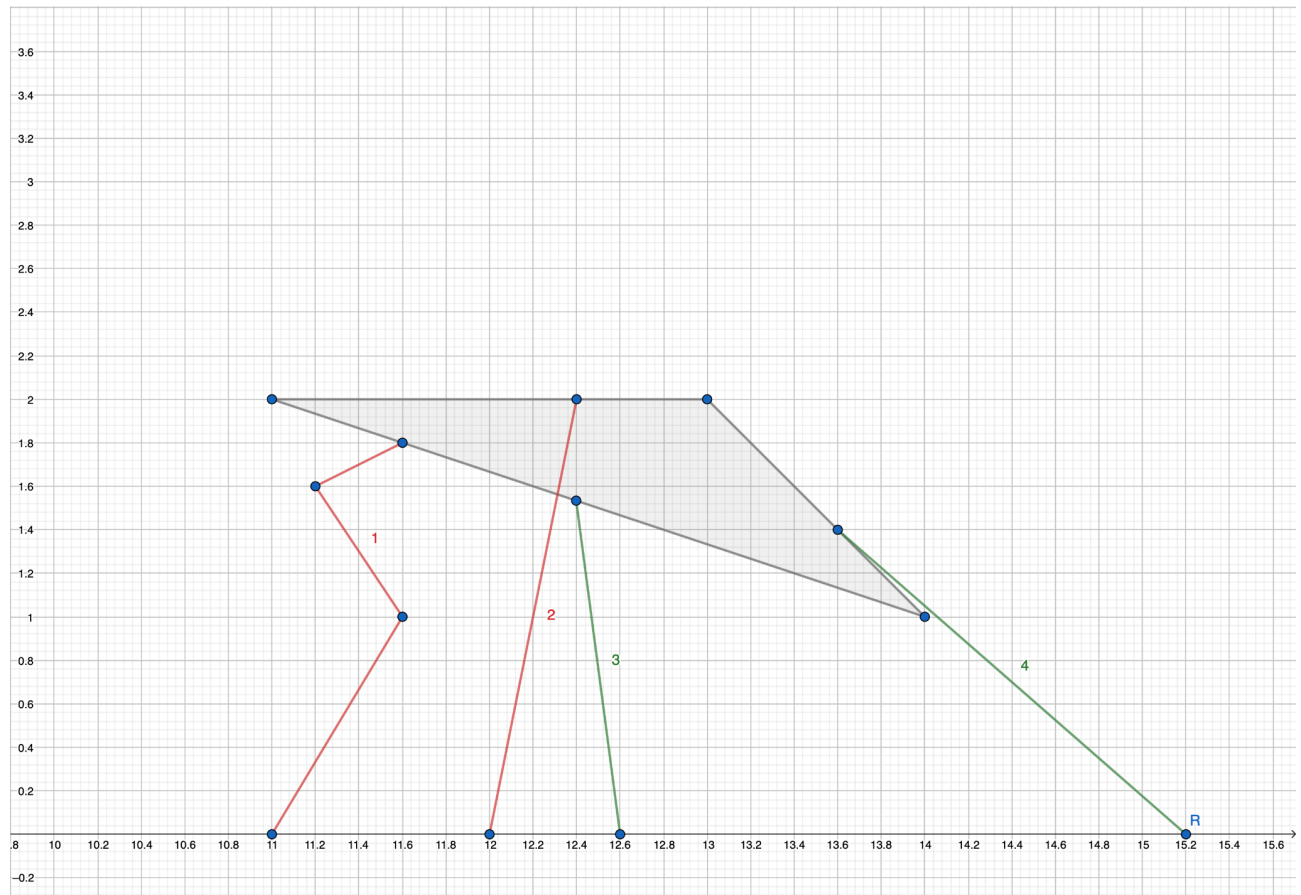


Reachable

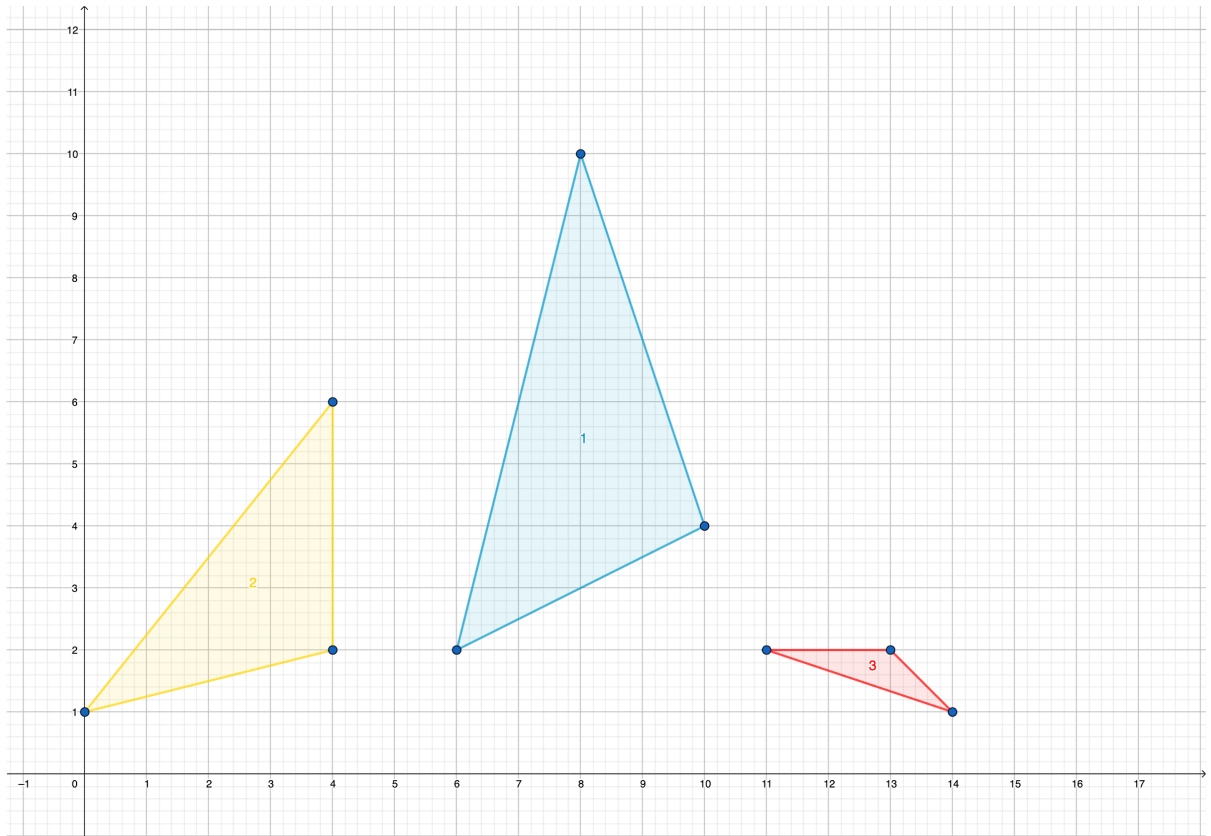
You are given a triangle whose coordinates are natural numbers. A point p , on the border of a triangle, is considered reachable if there is a straight line from point p to the x-axis which does not intersect with the triangle (excluding the border it lies on).



In the picture the triangle is marked with grey color. The first path is invalid because it does not go along a straight line, however, the point itself is reachable (given a straight line from the x-axis) The second path is invalid because it intersects with the interior of the triangle. The third and fourth paths are correct.

Find the total length of the unreachable parts on the triangle border from the x-axis. These parts are segments and their number is finite.

In the picture below, three cases are illustrated.



In the first two triangles, all points on the borders of the triangle are reachable, thus the answers are 0. In the third triangle (red), the answer is 2.

Given the following test cases and the triangles coordinates, submit your answers on the form:

Triangle 1:

(0, 0)

(4, 0)

(2, 4)

Triangle 2:

(0, 1)

(1, 1)

(0, 0)

Triangle 3:

(12, 3)

(4, 5)

(5, 6)

Triangle 4:

(3, 5)

(3, 4)

(7, 5)

Triangle 5:

(2, 6)

(9, 15)

(8, 14)